

PLATFORM COMPETITION IN TWO-SIDED MARKETS: EVIDENCE FROM THE PC VIDEO GAME MARKET



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Introduction

- Study of two-sided markets / network externalities in the PC game market
- Platforms (Steam and Epic) intermediate between sellers (publishers) and buyers (users), monetizing primarily through revenue sharing
- Low friction on both sides: publishers list at low cost and users pay no entry fee
- Epic entered the market in 2018 subsidizing both sellers (smaller revenue cut) and buyers (~\$2 billion in promotional spending)
- Did Epic's entry impact the dominant platform's (Steam) prices and sales?

Industry Background: How the PC Game Market Works

- PC gaming accounts for 20% of the videogame industry (\$190B industry) with game launchers like Steam / Epic accounting for 90% of the PC game market
- Unlike game consoles, PCs have no hardware gatekeeper dictating which games can be sold or not. Instead, anyone can make / sell a videogame on PC
- Game launchers serve as an online storefront for all participating games, while also handling game updates, multiplayer features, piracy prevention (DRM), and many more features

Industry Background: Publishers, Prices, and Steam's Position

- Publishers control prices and discounts, not the platform
- Publishers are not restricted to listing on one platform, many publish on both Steam and Epic
- Discounting / sales are a common practice in this market, and Steam / Epic encourage publishers to participate in seasonal sales that also serve as promotional events
- Before Epic launched their platform, Steam spent 10+ years as the only major game launcher / PC game storefront that sold games from multiple different publishers

Industry Background: Platform Comparison

Category	Steam	Epic
Launch Date	September 12, 2003	December 6, 2018
Monthly Active Users (2025 Avg)	132M	67M
Registered Accounts	N/A	317M
Revenue (2025)	\$16.9B	\$1.16B
Market Share	74-75%	~3% (by revenue share)
Games in Catalog	117,000+	6,000+
Revenue Cut	30% (tiered: 25% above \$10M; 20% above \$50M)	12%

Industry Background: Epic's Entry

- Epic enters the market in Dec 2018 as the first major competitor to Steam
 - Other platforms existed, but were primarily publisher-specific
- Epic attempted to capture market share through the buyer- and seller-side:
 - Buyer-Side: Weekly free games promotion, platform-exclusive titles, and Epic Rewards (cash back program)
 - Seller-Side: Significant revenue sharing cut (12% vs 30%), and 0% revenue sharing on first \$1M (implemented after study period in June 2025)
- Epic has grown as the second largest platform in the PC game market, but still falls behind Steam on all metrics
- Potentially in response to Epic, Steam updated their revenue sharing program November 30, 2018 to benefit larger publishers (flat 30% vs tiered system)

Literature Review: Foundation of Two-Sided Market Theory

Katz & Shapiro (1985):

- Foundational paper on network externalities (the utility of a good increases with the number of users)
- Dominance is self-fulfilling in markets with network externalities. Expected market leaders become market leaders.

Rochet & Tirole (2003):

- Platforms must choose a price structure, not just a price level, that incentivizes both buyers and sellers
 - One side is subsidized and typically becomes a loss leader
- Create the single-homing index (measure of how many users single-home their usage)
- When multihoming exists on one side, platforms attempt to “steer” users on the other side from competing platforms

Literature Review: Foundation of Two-Sided Market Theory

Rochet & Tirole (2006):

- Membership vs Usage: A user multi-homing registration across platforms does not imply that their usage will multi-home across those platforms.
 - When users single-home in terms of usage, it becomes harder for rival platforms to “steer” users with price competition
- Structure vs Level: Distinguishes a platform’s price level (total extracted from both sides) and price structure (how that total is divided between groups), and determines that a “two-sided market” is one where the price structure affects transaction volume independently of the price level
 - A platform’s competitive and pricing decisions on one side are distinct from those on the other side, despite being set jointly

Armstrong (2006):

- There are three factors that determine platform pricing: relative cross-group externalities, fixed fees vs per-transaction fees, and single-homing vs multi-homing
- Competitive Bottlenecks Model: A two-way market model where one side single-homes and the other multi-homes, giving the platform a local monopoly over its single-homing users

Literature Review: Dominance, Entry, and Welfare

Ko & Shen (2021):

- Market structure is determined by network-effect strength (β)
 - Weak Effects ($\beta \leq 0.5$) = symmetric equilibrium
 - Moderate Effects ($0.5 < \beta < 1$) = asymmetric equilibrium with a dominant platform
 - Strong Effects ($\beta \geq 1$) = monopoly equilibrium
- An asymmetric equilibrium with a dominant platform yields higher consumer surplus, producer surplus, and total welfare
 - A monopoly equilibrium always raises producer surplus and total welfare, but consumer surplus only rises when $1 \leq \beta < 1.1$

Ko & Shen - Online Appendix (2021):

- Results are robust to platforms charging a revenue share and sellers multi-homing, assuming buyers single-home

Literature Review: Dominance, Entry, and Welfare

Caillaud & Jullien (2003):

- Entrants to two-sided markets must use divide & conquer strategies to obtain market share
 - Subsidize one side's participation (divide) and recover the loss on the other side (conquer)
- Under multi-homing, it's easy for the entrant to obtain user membership (divide), but it's hard to steal the transaction (conquer) from the dominant platform.
- The incumbent obtains the bulk of transactions as the “first source”, while the entrant (“second source”) only captures what the incumbent misses

Literature Review: Same-Market Empirical Evidence

Jeong et al. (2026):

- Studies the impact of Epic's weekly free game promotional event on Steam revenue of the same games
 - Steam sales of the same game grew 59% over the promotion week, despite Epic offering the game for free
 - Over a ± 1 year window around treatment, Epic's promotional events raise Steam's price elasticity and induce more frequent and deeper sales
- Survey Findings:
 - The key driver is network externalities, constrained by price
 - Multiplayer features and availability of additional content was the biggest factor for obtaining the game on Steam over Epic

Market Characterization

- The PC game market resembles a two-sided market with moderate-to-strong network effects.
- The market sits in an asymmetric equilibrium where Steam is the dominant, “first source” platform and Epic is the “second source.”
- Sellers (Publishers) either multi-home between platforms, or have easy, low-cost access to multihoming.
- Buyers (Users) primarily membership-multi-home, but single-home on Steam in terms of usage.

H1: Short-Run Incumbent Insulation

H1: A game listing on Epic has no effect on that same title's Steam price, sale frequency, or discount depth in the short run.

Academic Justification:

- Katz & Shapiro (1985): Steam's dominance is self-fulfilling by its users.
- Armstrong (2006): Under competitive-bottleneck, platforms have a local monopoly over their single-homing users. This insulates buyer-side pricing.
- Rochet & Tirole (2003 & 2006): Price discipline operates through steering, which requires the multi-home side (publishers) to “steer” transactions to the cheaper platform, but steering is weak due to the single-homing usage on Steam.

H1: Short-Run Incumbent Insulation

H1: A game listing on Epic has no effect on that same title's Steam price, sale frequency, or discount depth in the short run.

Academic Justification:

- Caillaud & Jullien (2003): Under multi-homing, an entrant can divide (Epic's promotional activities claiming users), but not conquer (capture transactions / usage).
- Ko & Shen (2021): In a high β environment (β = network effect parameter), dominance by one platform is the equilibrium.
- Jeong et al. (2026): Their results shows that Steam was not negatively affected by Epic's promotional activities.

H2: Selection-Induced Differences

H2: Differences between Steam-Only and Dual-Platform games' prices and sales reflect selection into listing, not a causal effect of listing.

Academic Justification:

- Jeong et al. (2026): States that Epic selects high-potential games for promotion events and their catalog.
- Ko & Shen (2021): Sellers (Publishers) in an asymmetric equilibrium differ across platforms. This composition is endogenous to network effects.
- Caillaud & Jullien (2003): Divide and conquer entrants target particular seller compositions via subsidization, leading to selection into listing.

H3: Long-Run Pricing Drift

H3: Steam's prices and sale behavior drift over the long run in a direction consistent with sustained competitive pressure from Epic.

Academic Justification:

- Jeong et al. (2026): Found that Epic's promotional events raised Steam's price elasticity, while increasing sale frequency and discount depth for identical titles over a 2 year window.
- Caillaud & Jullien (2003) and Armstrong (2006): Their results on incumbent insulation depend on the incumbent retaining its local monopoly on usage-single-homing users, but Epic's consistent contestability and competitive pressure over a larger time horizon can erode the "single-homing" nature of Steam users over time, leading to potential downward pressure on prices.

Data

- All game data is sourced from the IsThereAnyDeal API, which tracks PC video game prices across all platforms / stores.
- Data Sample:
 - 900 games' daily price and sale data from 2013 to 2024
 - Games selected from “Most Popular”, “Most Collected”, and “Most Waitlisted” categories
 - Release years between 2013 - 2023
- CPI data was collected to inflation-adjust prices (base = 2015)
- Outcome Variables: Price, Real Price, Sale Frequency, and Discount Depth

Descriptive Statistics: Outcome Variables

Table 1: Outcome Variable Comparison by Group

Variable	Steam-Only	Dual-Platform	Difference
Pre-Epic Period (before Dec 2018)			
Avg Price (USD)	23.76	28.20	4.45***
Avg Real Price (USD)	22.90	27.19	4.29***
Sale Frequency	0.12	0.15	0.02**
Discount Depth (%)	5.57	5.65	0.08
N (games)	335	174	
Post-Epic Period (after Dec 2018)			
Avg Price (USD)	24.05	27.71	3.66***
Avg Real Price (USD)	19.95	22.63	2.68***
Sale Frequency	0.19	0.22	0.03***
Discount Depth (%)	10.24	11.85	1.62***
N (games)	544	356	
Full Sample Period			
Avg Price (USD)	24.56	28.41	3.85***
Avg Real Price (USD)	20.85	23.66	2.82***
Sale Frequency	0.18	0.21	0.03***
Discount Depth (%)	9.24	10.83	1.58***
N (games)	544	356	

- ~\$4.50 difference between game type before Epic
- Significant differences for all outcomes (except discount depth pre-Epic)
- Method: Average of each game's average

Descriptive Statistics: Release Year

Table 2: Release Year Distribution by Group

Release Year	Steam-Only (%)	Dual-Platform (%)	Difference
2013	0.6%	0%	-0.6
2014	11.4%	6.5%	-4.9**
2015	12.5%	8.4%	-4.1*
2016	11.6%	9.6%	-2
2017	10.7%	12.1%	1.4
2018	11.4%	10.4%	-1
2019	9.2%	9%	-0.2
2020	6.6%	9.8%	3.2
2021	7.7%	11.8%	4.1*
2022	9%	10.7%	1.7
2023	9.4%	11.8%	2.4

- Dual Platform games are skewed towards more recent release dates
- Only three years have significant differences

Descriptive Statistics: Genres

Table 3: Genre Distribution by Group

Genre	Steam-Only (%)	Dual-Platform (%)	Difference
Adventure	21.3%	31.2%	9.9***
OpenWorld	16.9%	21.9%	5*
RPG	30.5%	25.6%	-5
Narrative	15.4%	20.2%	4.8*
Multiplayer	20.2%	17.7%	-2.5
Survival	11.2%	13.5%	2.3
Platformer	9.6%	11.5%	2
Shooter	11%	9.3%	-1.8
Horror	8.1%	9.8%	1.7
Strategy	12.9%	14.6%	1.7
Puzzle	5.3%	6.7%	1.4
Action	32.2%	30.9%	-1.3
Simulation	13.1%	13.5%	0.4
Roguelike	7.2%	7%	-0.1

- Only three genres with significant differences
- Mostly fair balance of genres between the two groups
- Games can have multiple genres

Market-Level Analysis

Methodology - Section 1

Outcome Trend Graphs:

- Observe the trend in Steam prices and sales across the entire data sample
- Separated into game age windows (0-6mo, 6mo - 2yr, and 2yr+)

Post-Epic Launch Outcome Trend Group Comparison Graphs:

- Observes the trend in Steam prices and sales from 2018 - 2024 (Post-Epic)
- Compares Steam-Only, Dual-Platform, and Future Dual-Platforms trends
- Separated into game age windows (0-6mo, 6mo - 2yr, and 2yr+)

Methodology - Section 1

Model 1:

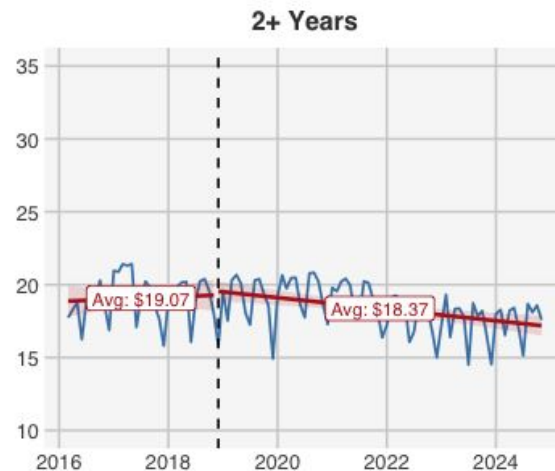
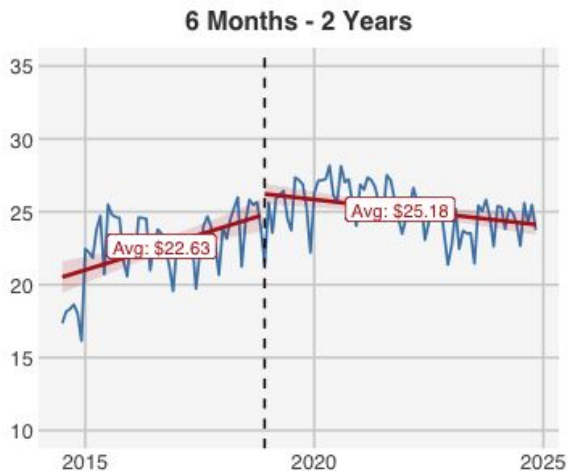
$$Y_{it} = \sum_{k \neq 2018} \beta_k \mathbb{1}[\text{year}_t = k] + \sum_{k \neq 2018} \delta_k (\mathbb{1}[\text{year}_t = k] \times \text{EverEpic}_i) + \gamma \log(\text{QSR}_{it}) + \alpha_i + \tau_{m(t)} + \varepsilon_{it}$$

- Year and Year:Ever-On-Epic variables (Ever-On-Epic is absorbed by game FE)
- Controls for time since release ($\log(\text{QSR})$)
- Game and month (of year) fixed effects
- Base year = 2018

Market-Level Analysis: Monthly Average Real Price on Steam

Monthly Average Real Price on Steam

By game age window | dashed = Epic launch



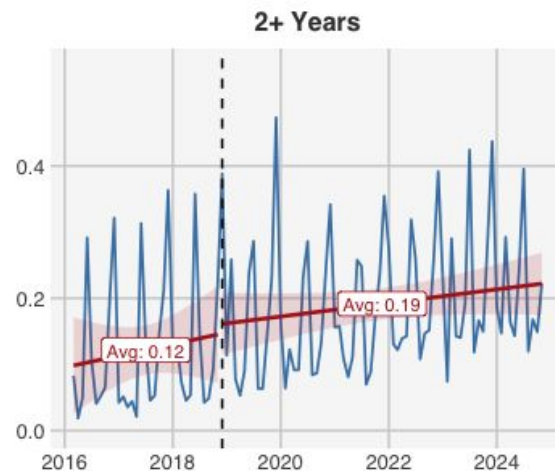
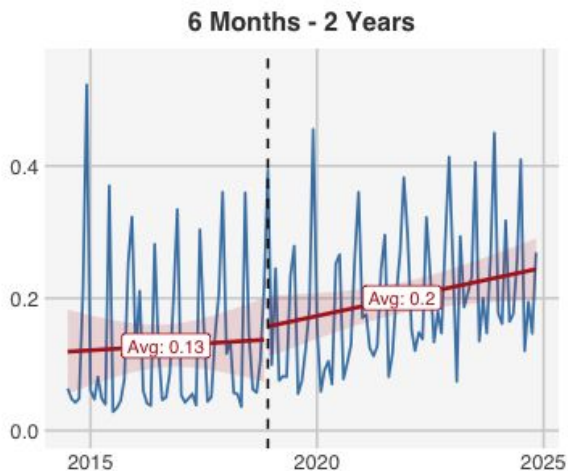
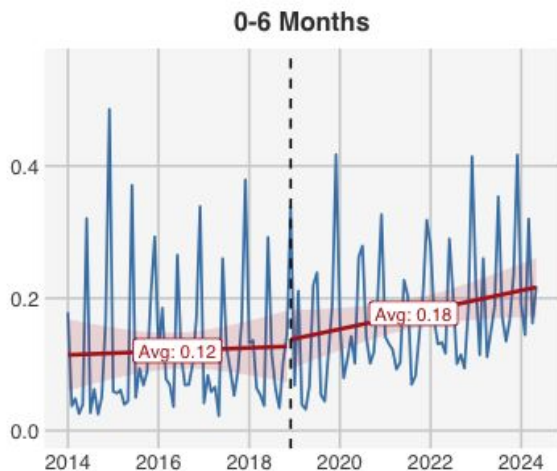
Time = Game Age Window

Dashed Line = Epic Launch

Market-Level Analysis: Monthly Average Sale Frequency on Steam

Monthly Average Sale Frequency on Steam

By game age window | dashed = Epic launch



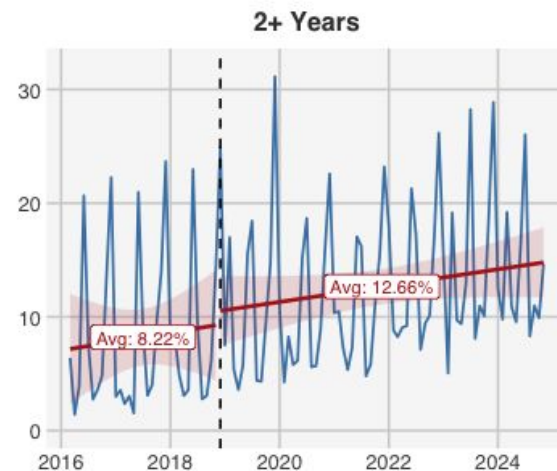
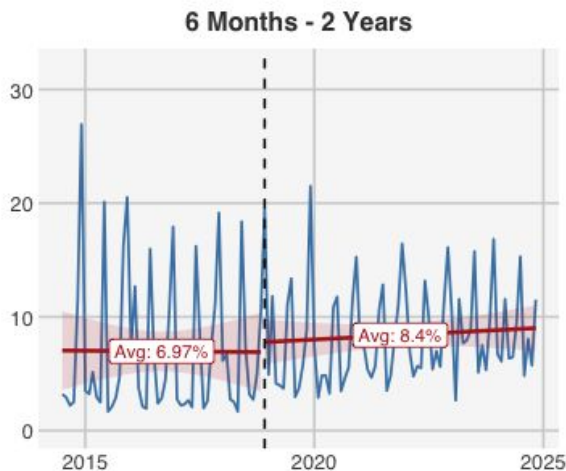
Time = Game Age Window

Dashed Line = Epic Launch

Market-Level Analysis: Monthly Average Discount Depth on Steam

Monthly Average Discount Depth on Steam

By game age window | dashed = Epic launch



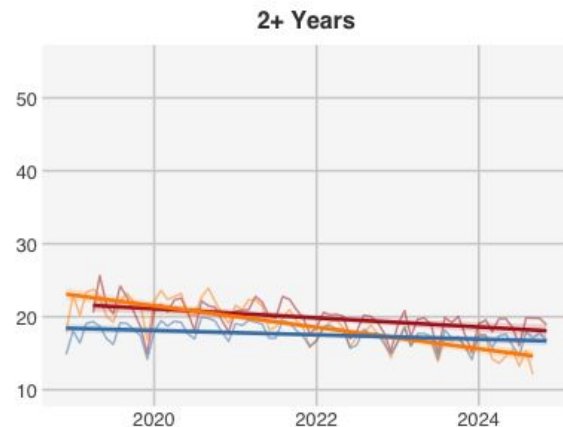
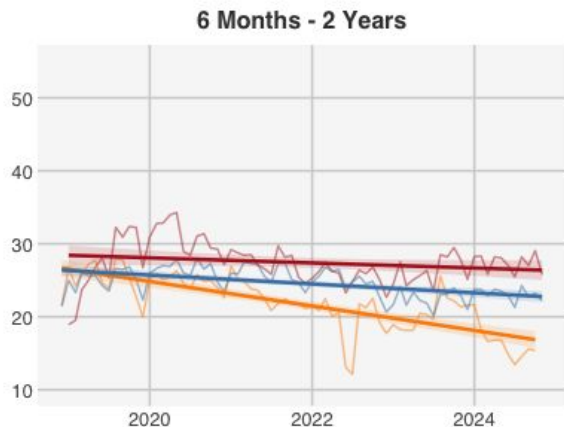
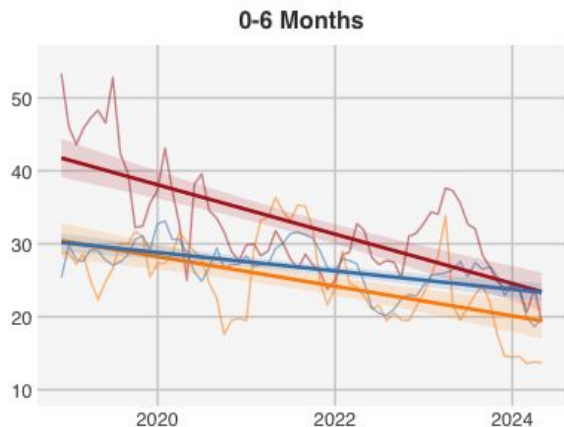
Time = Game Age Window

Dashed Line = Epic Launch

Market-Level Analysis: Post-Epic Launch Real Prices on Steam

Post-Launch Real Price on Steam

By Game Age Window | Trend lines with 95% CI



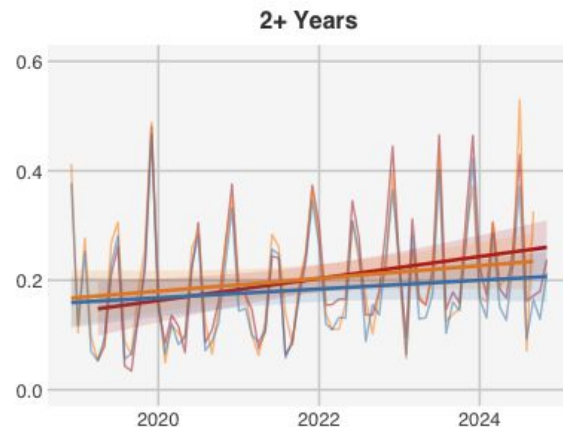
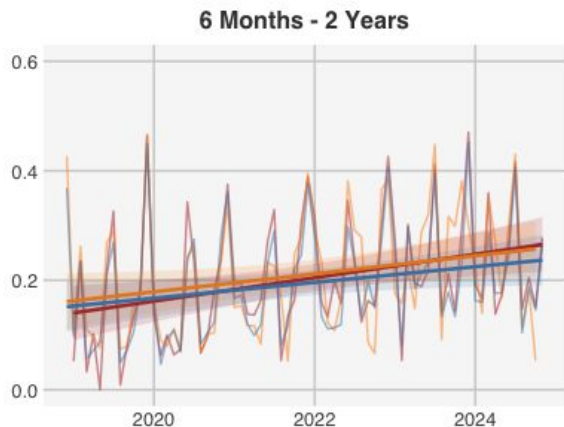
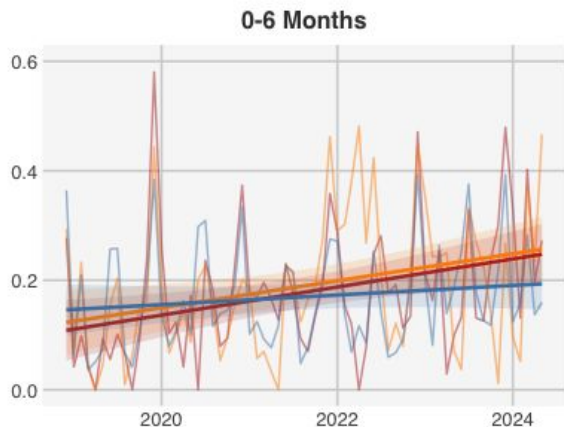
— Dual-Platform — Future Dual-Platform — Steam-Only

Time = Game Age Window

Market-Level Analysis: Post-Epic Launch Sale Frequency on Steam

Post-Launch Sale Frequency on Steam

By Game Age Window | Trend lines with 95% CI

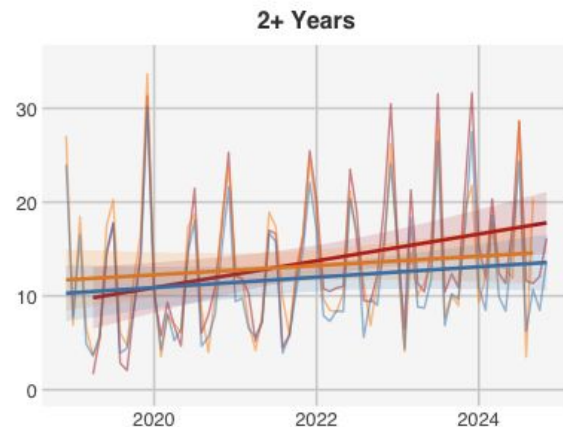
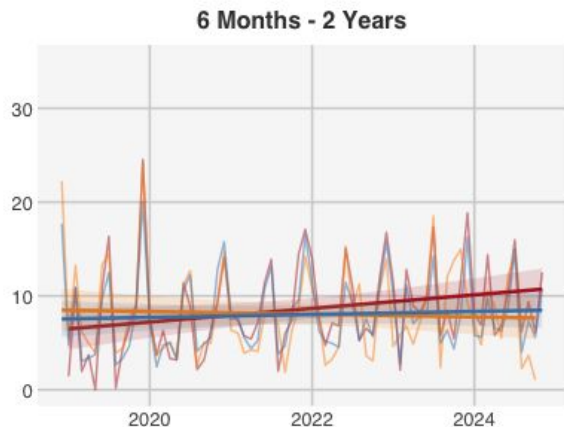
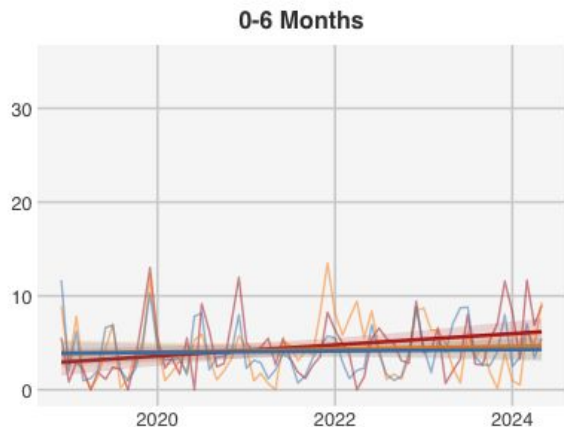


— Dual-Platform — Future Dual-Platform — Steam-Only

Time = Game Age Window

Market-Level Analysis: Post-Epic Launch Discount Depth on Steam

Post-Launch Discount Depth on Steam
By Game Age Window | Trend lines with 95% CI



Dual-Platform Future Dual-Platform Steam-Only

Time = Game Age Window

Market-Level Analysis: Regression Models

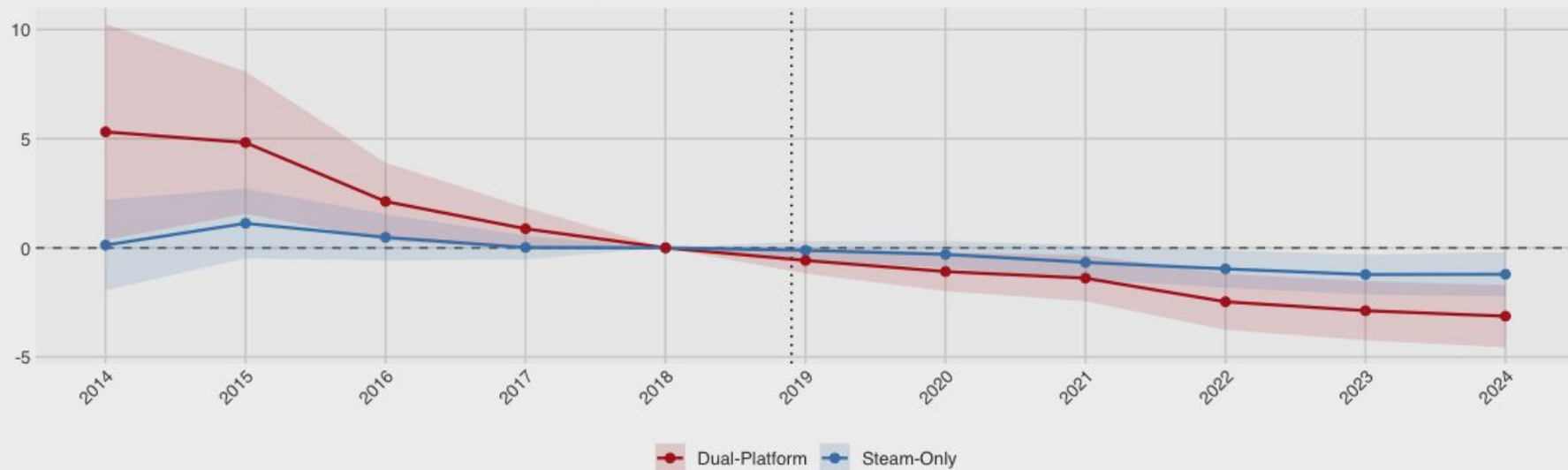
Year-Relative Effects on Steam Outcomes				
	Nominal Price (USD)	Real Price (USD)	Sale Freq. (share)	Discount Depth (pp)
2014	0.123	0.083	0.054**	2.557**
	(1.057)	(1.056)	(0.019)	(0.803)
2015	1.121	1.294	0.004	1.384*
	(0.816)	(0.818)	(0.010)	(0.603)
2016	0.480	0.770	-0.006	0.742*
	(0.540)	(0.537)	(0.006)	(0.354)
2017	0.016	0.235	-0.008	-0.072
	(0.278)	(0.272)	(0.004)	(0.222)
2019	-0.116	-0.208	0.021***	0.912**
	(0.216)	(0.207)	(0.005)	(0.297)
2020	-0.302	-0.393	0.011	0.081
	(0.313)	(0.302)	(0.006)	(0.351)
2021	-0.665	-1.291***	0.016*	0.231
	(0.396)	(0.383)	(0.006)	(0.404)
2022	-0.966*	-2.758***	0.038***	1.468**
	(0.438)	(0.425)	(0.008)	(0.508)
2023	-1.224**	-3.544***	0.059***	2.558***
	(0.466)	(0.458)	(0.009)	(0.593)
2024	-1.212*	-3.849***	0.066***	2.910***
	(0.517)	(0.508)	(0.009)	(0.633)
2014 x Epic	5.190*	5.426*	-0.044	-1.163
	(2.603)	(2.590)	(0.026)	(1.042)
2015 x Epic	3.699*	4.092*	-0.018	-0.609
	(1.756)	(1.763)	(0.014)	(0.773)
2016 x Epic	1.643	1.945	-0.001	-0.240
	(1.009)	(1.014)	(0.009)	(0.517)
2017 x Epic	0.858	0.953	0.004	-0.266

	Nominal Price (USD)	Real Price (USD)	Sale Freq. (share)	Discount Depth (pp)
	(0.553)	(0.545)	(0.009)	(0.327)
2019 x Epic	-0.462	-0.483	-0.003	0.466
	(0.341)	(0.328)	(0.007)	(0.394)
2020 x Epic	-0.787	-0.835	-0.003	0.319
	(0.508)	(0.490)	(0.008)	(0.472)
2021 x Epic	-0.726	-0.897	0.000	0.745
	(0.598)	(0.585)	(0.008)	(0.503)
2022 x Epic	-1.506*	-1.721*	0.016	2.026**
	(0.686)	(0.667)	(0.010)	(0.677)
2023 x Epic	-1.657*	-1.904**	0.012	2.316**
	(0.709)	(0.692)	(0.011)	(0.783)
2024 x Epic	-1.922**	-2.144**	0.021*	3.408***
	(0.714)	(0.703)	(0.010)	(0.756)
Log(QSR)	-1.219***	-1.682***	0.008	2.994***
	(0.252)	(0.247)	(0.004)	(0.207)
Num Obs.	69654	69654	69654	69654
R2 Within	0.128	0.309	0.037	0.128
* p < 0.05, ** p < 0.01, *** p < 0.001				
All models include game and month-of-year fixed effects. Reference year 2018. SEs clustered by game.				

Market-Level Analysis: Regression Output (Price)

Year-Relative Change in Nominal Price

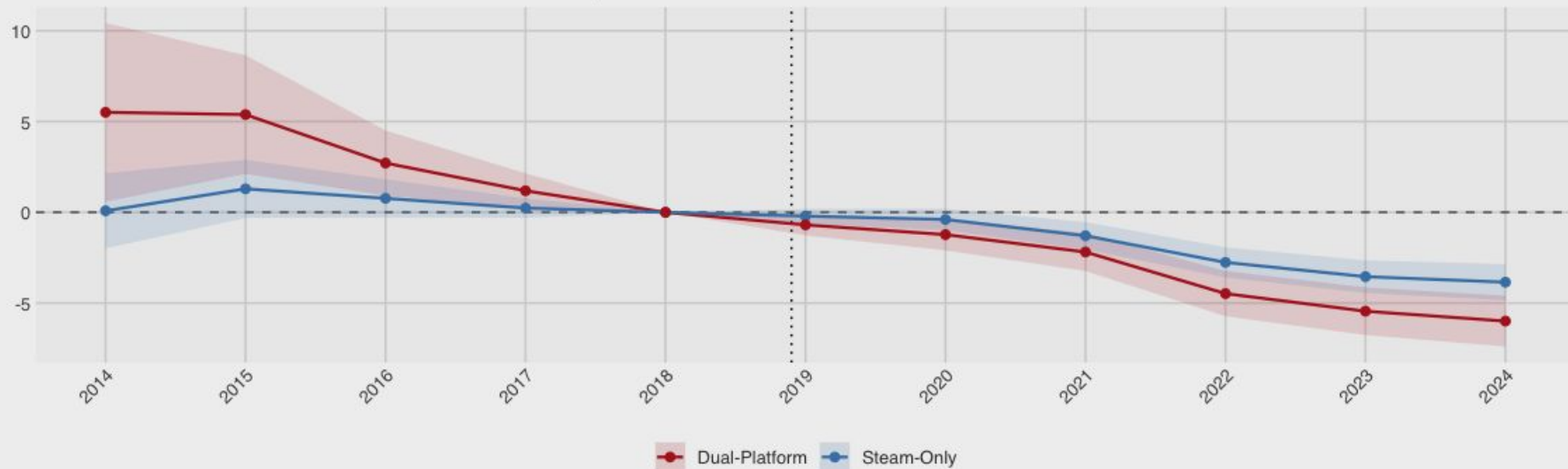
Ref = 2018 | Point estimate with 95% CI | dotted line = Epic launch



Market-Level Analysis: Regression Output (Real Price)

Year-Relative Change in Real Price

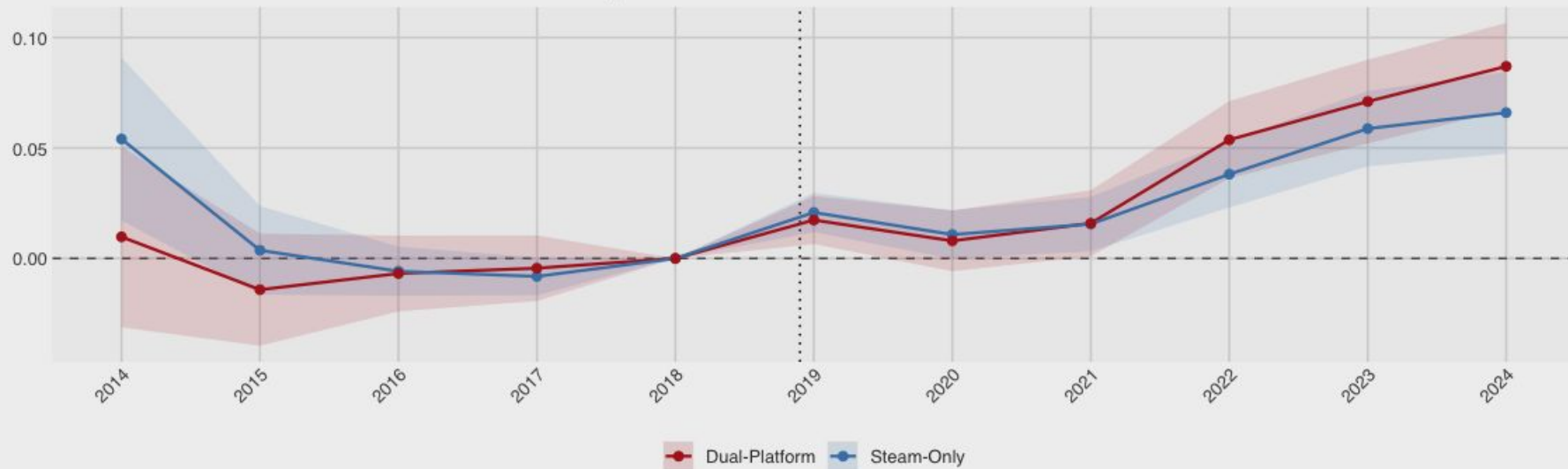
Ref = 2018 | Point estimate with 95% CI | dotted line = Epic launch



Market-Level Analysis: Regression Output (Sale Frequency)

Year-Relative Change in Sale Frequency

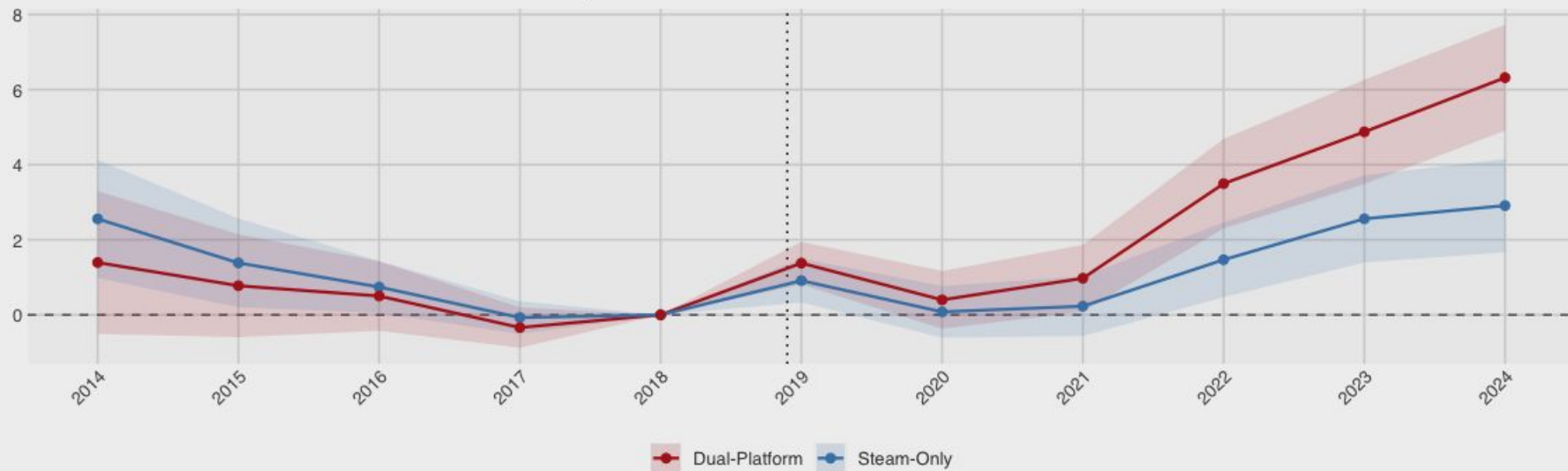
Ref = 2018 | Point estimate with 95% CI | dotted line = Epic launch



Market-Level Analysis: Regression Output (Discount Depth)

Year-Relative Change in Discount Depth

Ref = 2018 | Point estimate with 95% CI | dotted line = Epic launch



Game-Level Analysis

Methodology - Section 2

- Note: Treated = Game that joins Epic

T-Tests:

- Two-Sample t-tests performed on treated games to observe changes in outcome variables before and after listing on Epic
- Time Windows: 30d, 60d, 90d, and 180d before and after entry

Methodology - Section 2

Model 2:

$$Y_{it} = \beta \text{OnEpic}_{it} + \alpha_i + \lambda_w + \varepsilon_{it}$$

- OnEpic = 1 if game is listed on Epic
- Game and week (of year) fixed effects
- Time Windows: 30d, 60d, 90d, and 180d before and after entry

Methodology - Section 2

Model 3: Stacked DiD

$$Y_{dik} = \tau (\text{Treated}_i \times \text{Post}_k) + \alpha_{di} + \gamma_{dk} + \varepsilon_{dik}$$

- Treated = Game listed on Epic | Post = Time after game is listed on Epic
- Week and game fixed effects
- Each game has one Steam-Only matched control (for each outcome)
 - Matched on pre-period trend of outcome, price level, genre overlap, and release date
- Time Windows: 60d, 90d, and 180d before and after entry

Methodology - Section 2

Model 4: Synthetic Control DiD

$$Y_{dik} = \tau (\text{Treated}_i \times \text{Post}_k) + \alpha_{d, \text{Treated}_i} + \gamma_{dk} + \varepsilon_{dik}$$

- Same model, with synthetic controls
- Synthetic controls are a weighted combination of Steam-Only games that match pre-period trends
- Fits pre-trends better than 1:1 controls

Game-Level Analysis: T-Tests

T-Test Outcomes

Outcome	Window (days)	Rose	Fell	Rose ($p < 0.05$)	Fell ($p < 0.05$)
Price	30	105	162	65	68
Price	60	114	153	67	72
Price	90	117	150	72	80
Price	180	115	152	47	85
Sale Frequency	30	107	160	65	63
Sale Frequency	60	123	144	73	62
Sale Frequency	90	131	136	81	69
Sale Frequency	180	140	127	77	39
Discount Depth	30	105	162	66	63
Discount Depth	60	128	139	74	64
Discount Depth	90	135	132	83	67
Discount Depth	180	151	116	83	33

Price:

- More games fell in price
- Significance increased for “Fell” and decreased for “Rose”

Sale Frequency:

- More games increased sale frequency as window expanded
- Significance increased for “Rose” and decreased for “Fell”

Discount Depth:

- More games increased discount depth as window expanded
- Significance increased for “Rose” and decreased for “Fell”

Game-Level Analysis: Regression Models (Price)

Price ~ On Epic

	30d	60d	90d	180d
On Epic (post-entry)	-0.236 (0.447)	-0.183 (0.373)	0.039 (0.341)	0.180 (0.263)
Num Obs.	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- No significant results
- Within these time windows, games joining Epic does not affect the price of that game on Steam

Game-Level Analysis: Regression Models (Sale Frequency)

Sale Frequency ~ On Epic

	30d	60d	90d	180d
On Epic (post-entry)	0.023 (0.025)	-0.003 (0.021)	-0.002 (0.019)	-0.009 (0.014)
Num Obs.	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- No significant results
- Within these time windows, games joining Epic does not affect the frequency of sales for that game on Steam

Game-Level Analysis: Regression Models (Discount Depth)

Discount Depth ~ On Epic

	30d	60d	90d	180d
On Epic (post-entry)	0.201 (1.406)	0.137 (1.118)	-0.185 (0.996)	-0.157 (0.718)
Num Obs.	17817	34933	51731	101065
R2 Within	0.000	0.000	0.000	0.000

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

- No significant results
- Within these time windows, games joining Epic does not affect the size of discounts for that game on Steam

Game-Level Analysis: DiD Models (Pre-Treatment Fit)

Outcome	Window (days)	Joint pre-trend test (p-value)	
		Stacked DiD	Synthetic Control DiD
Price (USD)	60	0.212 ✓	0.402 ✓
	90	0.278 ✓	0.327 ✓
	180	0.161 ✓	0.008 X
Sale Frequency	60	0.811 ✓	0.101 ✓
	90	0.570 ✓	0.343 ✓
	180	0.567 ✓	0.057 ✓
Discount Depth (%)	60	0.535 ✓	0.368 ✓
	90	0.415 ✓	0.656 ✓
	180	0.357 ✓	0.165 ✓

H0: all pre-period event-study coefficients = 0 jointly (Wald chi-square, reference week = -1). High p-values support parallel trends. 17 of 18 tests pass.

- Joint Wald test that all pre-treatment coefficients equal 0
- A high p-value implies a flat pre-trend and implies passed parallel trends assumption
- All models pass, except 180 day synthetic control DiD on price

Game-Level Analysis: DiD Models (Price)

Effect of Epic Listing on Price (USD): Stacked vs. Synthetic Control DiD

	Stacked DiD			Synthetic Control DiD		
	60d	90d	180d	60d	90d	180d
ATT (treated x post)	0.499	0.475*	0.138	0.348	0.530*	0.233
	(0.277)	(0.231)	(0.199)	(0.248)	(0.207)	(0.196)
Num Obs	8596	12110	22986	8586	12104	22974

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Week and Game Fixed Effects

SE clustered by Game

Game-Level Analysis: DiD Models (Sale Frequency)

Effect of Epic Listing on Sale Frequency: Stacked vs. Synthetic Control DiD

	Stacked DiD			Synthetic Control DiD		
	60d	90d	180d	60d	90d	180d
ATT (treated x post)	-0.006	-0.002	0.005	0.009	0.009	0.018
	(0.011)	(0.009)	(0.008)	(0.012)	(0.009)	(0.010)
Num Obs	8596	12110	22986	8586	12104	22974
* p < 0.05, ** p < 0.01, *** p < 0.001						

Week and Game Fixed Effects

SE clustered by Game

Game-Level Analysis: DiD Models (Discount Depth)

Effect of Epic Listing on Discount Depth (%): Stacked vs. Synthetic Control DiD

	Stacked DiD			Synthetic Control DiD		
	60d	90d	180d	60d	90d	180d
ATT (treated x post)	-0.119	-0.048	0.356	0.500	0.349	1.186**
	(0.605)	(0.549)	(0.472)	(0.560)	(0.476)	(0.374)
Num Obs	8596	12110	22986	8586	12104	22974

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Week and Game Fixed Effects

SE clustered by Game

Summary of Findings

Market-Level Findings: When controlling for depreciation and game-specific qualities, both groups (Steam-Only and Dual-Platform) see small price declines with slightly larger and more frequent sales. These findings are stronger for Dual-Platform games, but none of the results are causal

Game-Level Findings: A game listing on Epic does not decrease the price of the same game on Steam. Rather, some results indicate that prices rise after 90 days of the listing (\$0.47 and \$0.53 ↑). Sale frequency remains unchanged, but discount depth increases by 1.19 pp after 180 days.

Hypothesis Evaluation

H1: A game listing on Epic has no effect on that same title's Steam price, sale frequency, or discount depth in the short run.

Findings:

- The stacked and synthetic control DiD show a slight increase in price (\$0.47 and \$0.53 ↑) in the 90 day window
- Only other significant result is discount depth (1.19) in 180 day window
- Games listing on Epic do not decrease the price of the same game on Steam, but can increase the price slightly in a 90 day window

Hypothesis Evaluation

H2: Differences between Steam-Only and Dual-Platform games' prices and sales reflect selection into listing, not a causal effect of listing.

Findings:

- Dual-Platform games were ~\$4.50 more expensive than Steam-Only games in the pre-Epic period.
- Dual-Platform games had slightly newer releases on average
- Jeong et al. (2026) states that Epic does not select games to their catalog randomly

Hypothesis Evaluation

H3: Steam's prices and sale behavior drift over the long run in a direction consistent with sustained competitive pressure from Epic.

Findings:

- A few years after Epic's release, games' Steam prices fall slightly, while sale frequency and discount depth increased.
- Results are stronger for games included on both platforms
- Insignificant results for first few years of post-launch period
- Results are not causal

Discussion

H1: In the short-run, a game listing on Epic can effect that game's Steam listing. Game prices increase slightly in a 90 day window, and discount depth increases in a 180 day window

H2: Based on initial price differences and the findings in Jeong et al. (2026), Epic's selection of games for its catalog was not random

H3: Based on the market-level results, prices and sales have changed over the time sample. Results aren't causal, but could indicate Epic slowly stealing single-homing users.

Future Research

- The primary limitation of this study is the lack of quantity sold / revenue data from these Steam games. With this data, researchers could dive deeper into if Epic impacted Steam revenue and if there's major differences between games that only list on Steam or list on both platforms.
- Additionally, differentiating games based on network externalities (ex: multiplayer features) may show the exact features making users single-home on Steam.
- Adjustments in network externalities in the sample period

Conclusion

The PC game market supports two-sided market / network externality literature:

- The market has an asymmetric equilibrium with Steam as the dominant, “first source” platform
- In the short-run, Epic launching and listing games exerts no downward pressure on Steam prices
 - Sale frequency remains unchanged, while discount depth only rises in the largest horizon tested
- In the long-run, market-level trends are consistent with competitive pressure eroding Steam’s local monopoly on its users, but the design cannot establish causality